PYTHON · CUSTOMTKINTER

Muskan's Ultimate Calculator

A dual-mode scientific calculator with real-time history, keyboard support & live clock

Standard Mode

Scientific Mode

History Log

Made with Python & CustomTkinter

KEY FEATURES

What makes this calculator special

Dual Mode Tabs

Standard & Scientific modes in separate tabs. Easily switch between everyday math and advanced calculations.

History Panel

Every calculation is logged in a live history panel. Clear it anytime with one click.

Keyboard Support

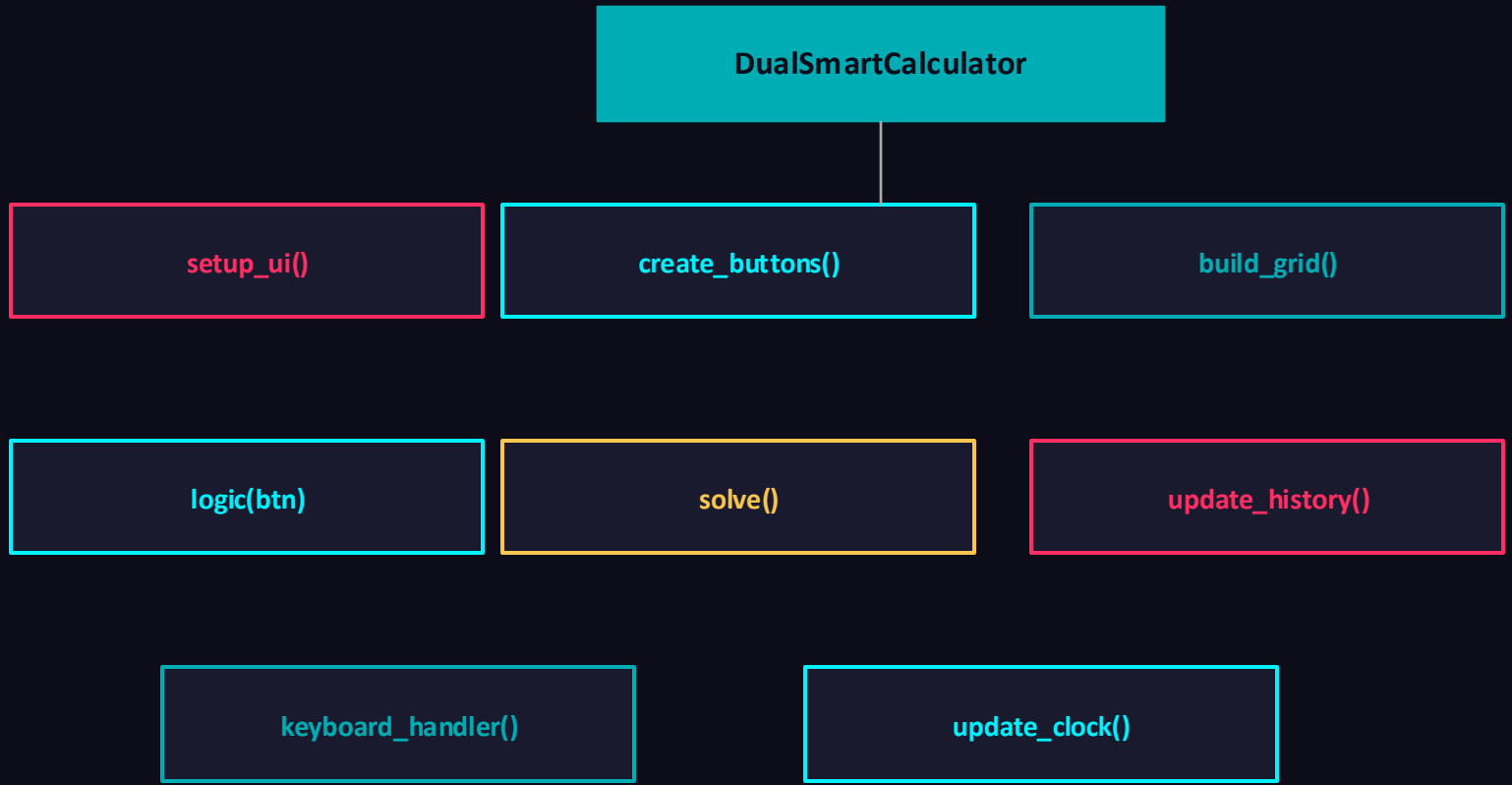
Full keyboard handler for digits, operators, Enter to solve, Backspace to delete, and Escape to clear.

Live Clock

A real-time HH:MM:SS clock is displayed in the top bar and updates every second.

CODE ARCHITECTURE

How the app is structured



SCIENTIFIC FUNCTIONS

Supported Mathematical Operations

Trigonometry

sin(x)
cos(x)
tan(x)
DEG / RAD modes
Undefined check at 90°/270°

Logarithms

log(x) → log base 10
ln(x) → natural log
Error on non-positive input

Powers & Roots

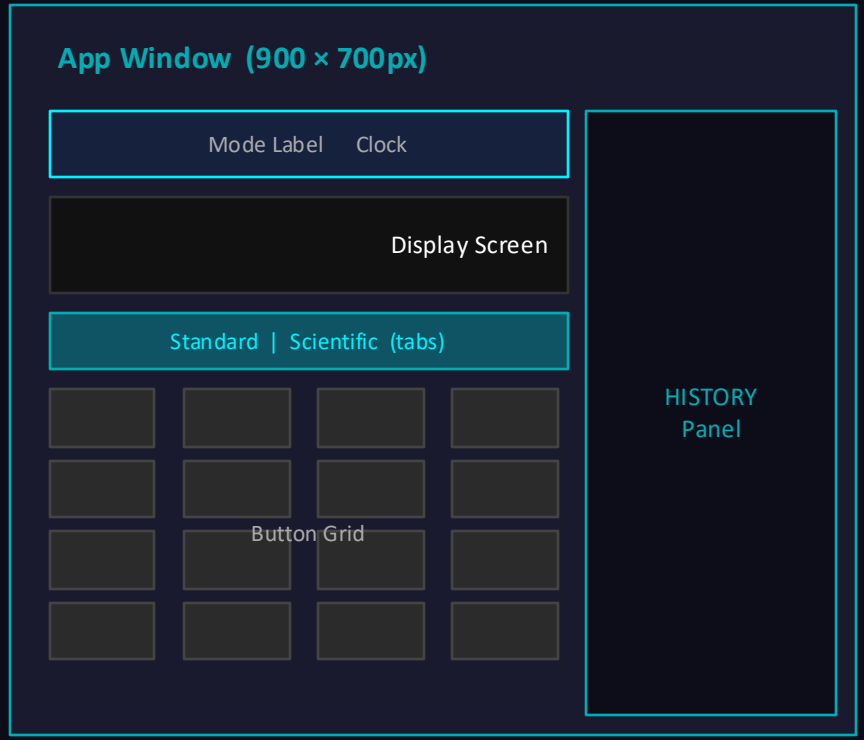
x^2 x^3 x^y
√x (square root)
EXP notation
ABS (absolute value)

Constants

π (pi = 3.14159...)
e (Euler = 2.71828...)
fact(n) factorial

UI DESIGN

Interface Layout & Design Choices



Design Choices

- Dark theme using CustomTkinter dark mode
- 900×700 fixed window, no resize
- Teal (#00ADB5) as primary accent color
- Pink/red (#FF2E63) for destructive actions (C, ⌫)
- CTkTabview for Standard / Scientific tabs
- CTkTextbox for scrollable history log
- Arial Bold throughout for clarity

HOW IT WORKS

Execution Flow

1

User Input

Button click or keyboard event triggers logic(btn)

2

Expression Built

Characters appended to self.expression string

3

Solve Called

eval() executes expression in a safe context dict

4

Result Shown

Result displayed on screen via StringVar

5

History Updated

Entry added to scrollable CTkTextbox history

⚠️ eval() is sandboxed with {"__builtins__": None} and a safe context — preventing code injection

SUMMARY

Muskan's Ultimate Calculator

— delivers a complete, real-world calculator experience —

- ✅ Dual-tab interface: Standard & Scientific
- ✅ Trigonometry with DEG/RAD switching & edge-case handling
- ✅ Safe expression evaluation with sandboxed eval()
- ✅ Full keyboard control: digits, operators, Enter, Backspace
- ✅ Persistent session history with clear button
- ✅ Live real-time clock in the header

Thank You!